Estimating Cosmological Parameters from the Matter Power Spectrum

Amortized simulation-based inference with neural posterior estimation

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Simulation-Based Inference (SoSe26) · TU Dortmund · Supervisor: Aayush

$\theta = \{ H_0 , \Omega_m , n_s , A_s \} \leftarrow P(k)$

Why this problem matters

The Universe is not smooth — matter clumps into the cosmic web.

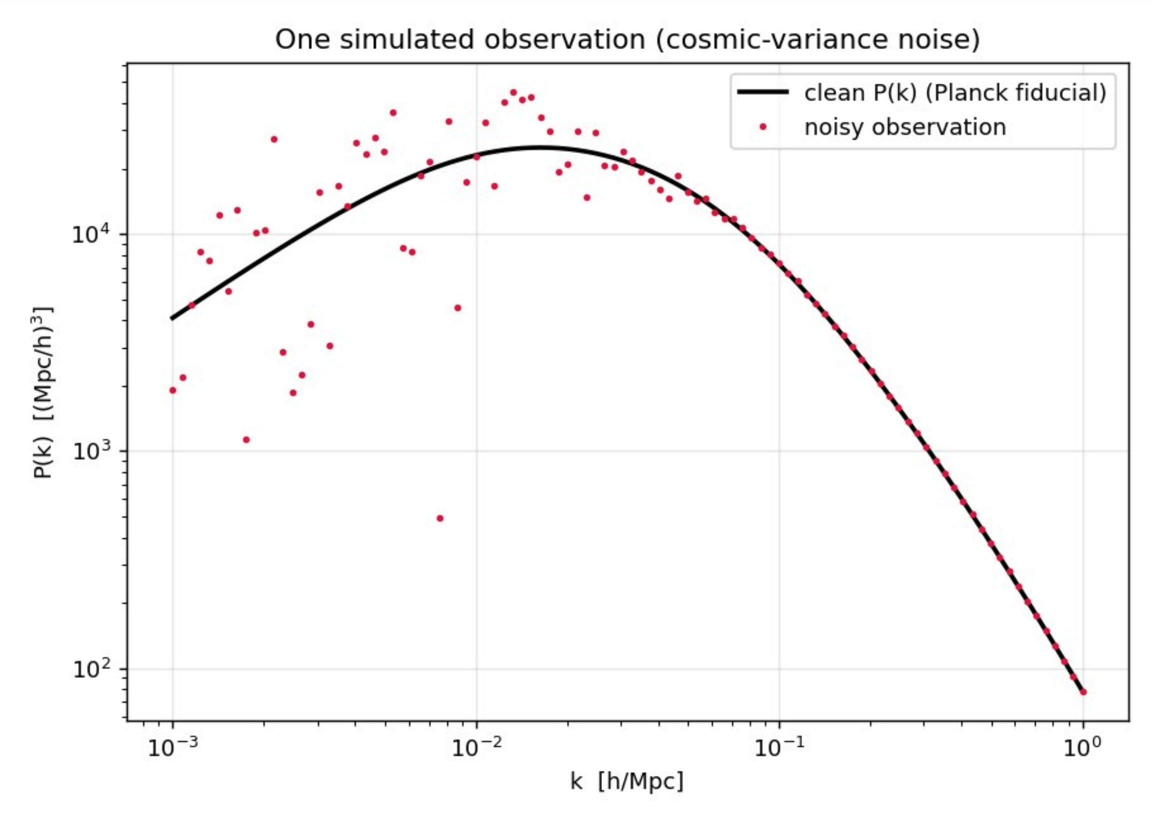
The matter power spectrum P(k) measures how much structure exists at each scale; galaxy surveys (DESI, Euclid) measure it.

The inverse problem:

We can simulate parameters → P(k) (forward), but we want P(k) → parameters, with honest uncertainty (backward).

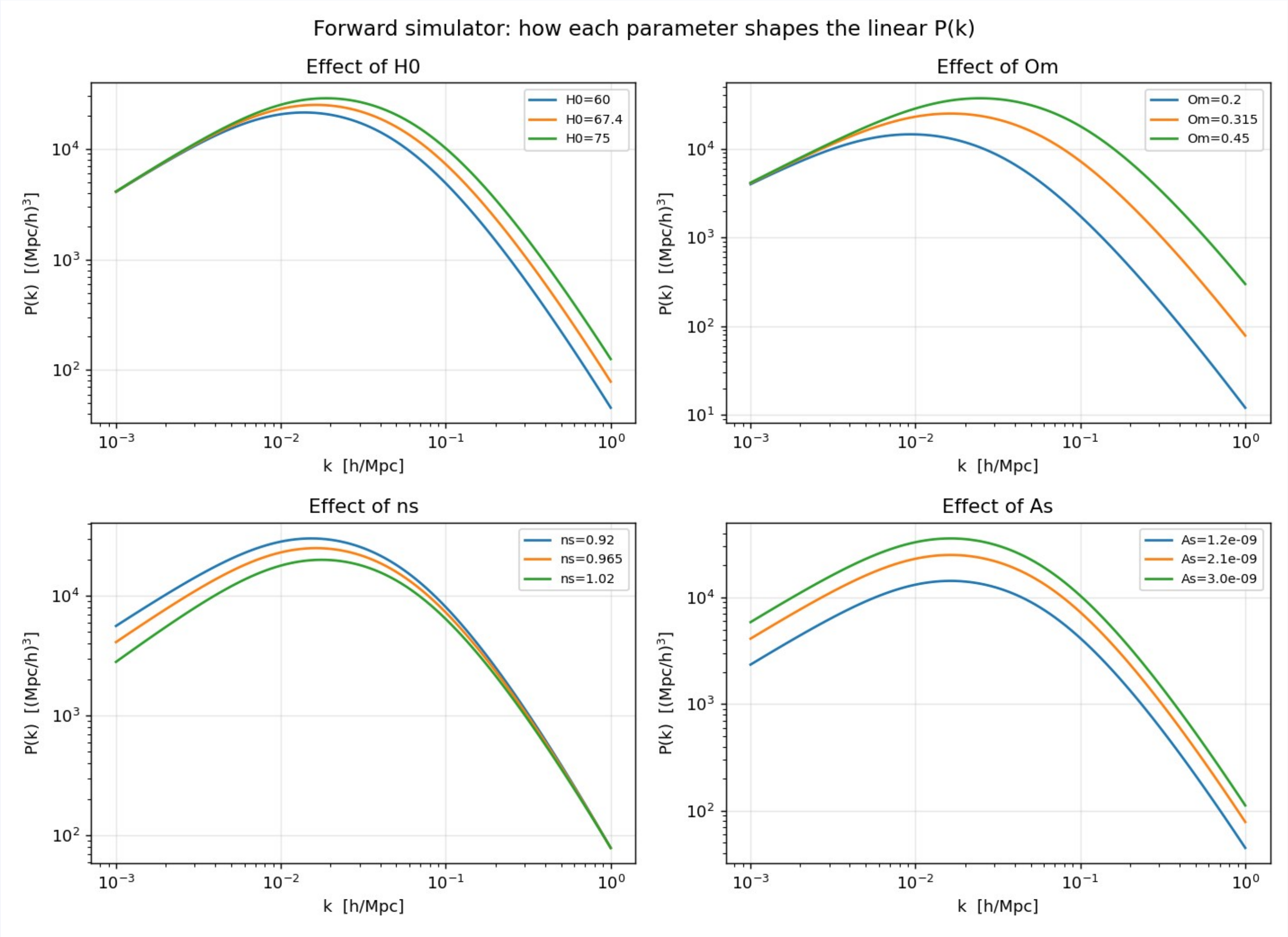
Why SBI:

- Realistic simulators have no tractable likelihood.
- Amortized inference: train once, infer any new spectrum instantly.
- Calibrated error bars → trustworthy cosmology (e.g. the Hubble tension).



One noisy observation of P(k) (cosmic-variance noise)

The four parameters shape P(k)



Each parameter leaves a distinct imprint on the spectrum

A_s amplitude
vertical scaling of P(k)

n_s spectral index
tilts / pivots the spectrum

Ω_m matter density
moves the turnover scale

H_0 Hubble constant
weak effect — enters only via shape → hard to constrain alone

Method: amortized neural posterior estimation

Prior
$\theta = \{H_0, \Omega_m, n_s, A_s\}$

→

Simulator
CAMB / BBKS + noise

→

P(k)
noisy spectrum

Train once on many simulations →
inference for any new spectrum is one fast forward pass (amortized).

Neural Posterior
$q(\theta \mid P(k))$

→

posterior
over θ

Built with: CAMB / BBKS simulator · Gaussian NPE (Python, NumPy, autograd) · BayesFlow for production · Go + HTML live demo

Likelihood-free: we only need to simulate, never to write $p(\text{data} \mid \theta)$. Maps to course Ch10 (ABC) → Ch12 (NPE/BayesFlow).

Simulator, priors & noise

Forward model

Linear $P(k)$ at $z=0$ on 100 log-spaced $k \in [10^{-3}, 1]$ h/Mpc.
Prototype: BBKS+Sugiyama transfer fn (≈ CAMB); production: CAMB.

Priors (proper, uniform)

$H_0 \sim U(60, 80)$ km/s/Mpc
$\Omega_m \sim U(0.10, 0.50)$
$n_s \sim U(0.90, 1.05)$
$\ln(10^{10} A_s) \sim U(2.5, 3.5)$ (log-amplitude)
fixed $\omega_b = 0.0224$, ω_c derived

Cosmic-variance noise

$\sigma_i = P(k_i) \sqrt{(2 / N_i)}$
$N_i = k_i^2 \, \Delta k \, V / (2\pi^2)$, $V = 10^9$ (Mpc/h)^3
Few modes at low k → noisy; many at high k → precise.

Preprocessing

$\log_{10} P(k)$, then standardize (raw P spans orders of magnitude).

Data: 12k train / 2k val / 3k test; fresh noise each batch.

Network & training

Architecture (≈31k params)

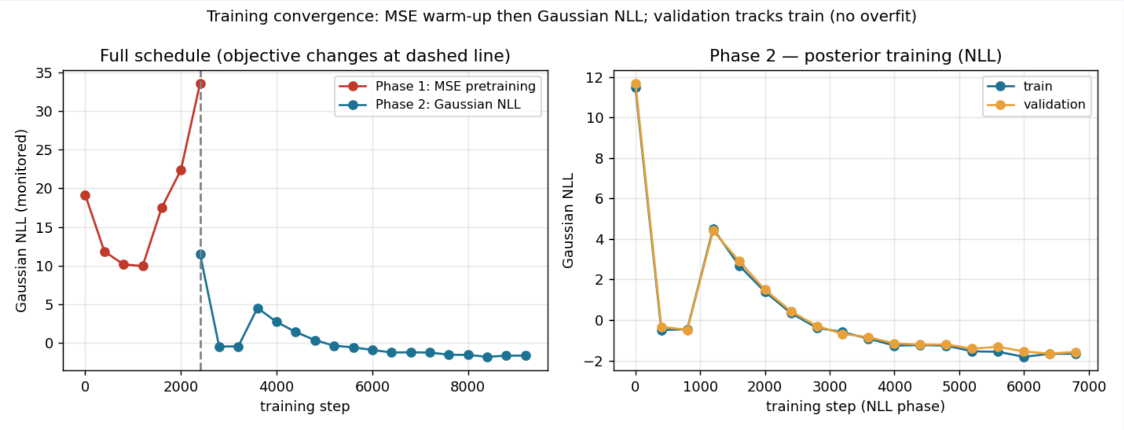
- Input: standardized log P(k) — 100 values
- MLP: 100 → 128 → 128, tanh (hidden layers = summary)
- Output 14 = mean(4) + Cholesky(10) → 4-D Gaussian, full covariance
- No dropout / weight decay — online noise acts as regularization

Training

- Loss: Gaussian negative log-likelihood (NLL)
- Two-phase: MSE warm-up (2.5k) → NLL (7k steps)
- Adam, batch 256, step LR decay (2e-3 → 1e-3)
- Offline sims + fresh online noise per batch
- ~20 s on CPU; val tracks train → no overfit

Tools: Python · NumPy · autograd

Production → CNN summary + BayesFlow normalizing flow

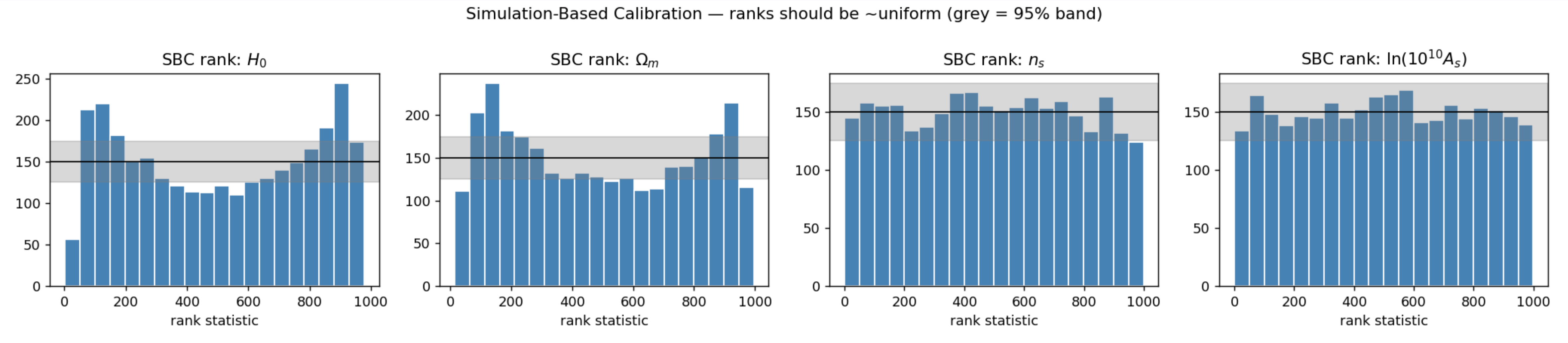


Training / validation convergence

Network internals (forward pass)

P(k) 100 → **Dense 128** tanh → **Dense 128** tanh → **14 outputs** mean + Cholesky → **4-D Gaussian** posterior

Calibration — is the posterior trustworthy?



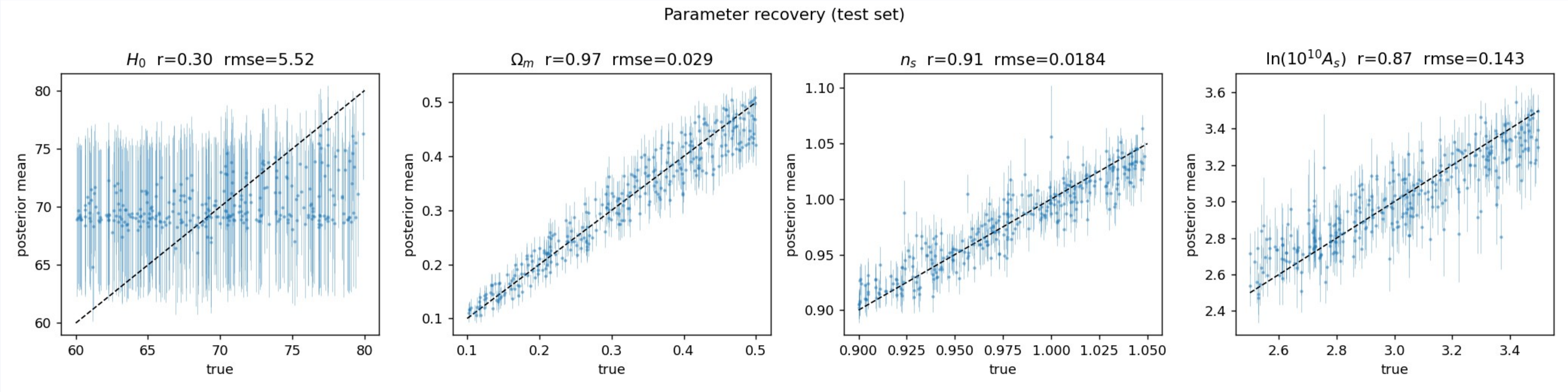
n_s, A_s ranks flat → well calibrated.

Ω_m, H_0 mild ∪-shape → slight tail overconfidence (typical for a Gaussian posterior; a normalizing flow would improve it).

z-score std ≈ 0.95, 95% coverage ≈ 0.96–1.00.

Contraction high for Ω_m/n_s/A_s, ≈ 0 for H_0.

Parameter recovery

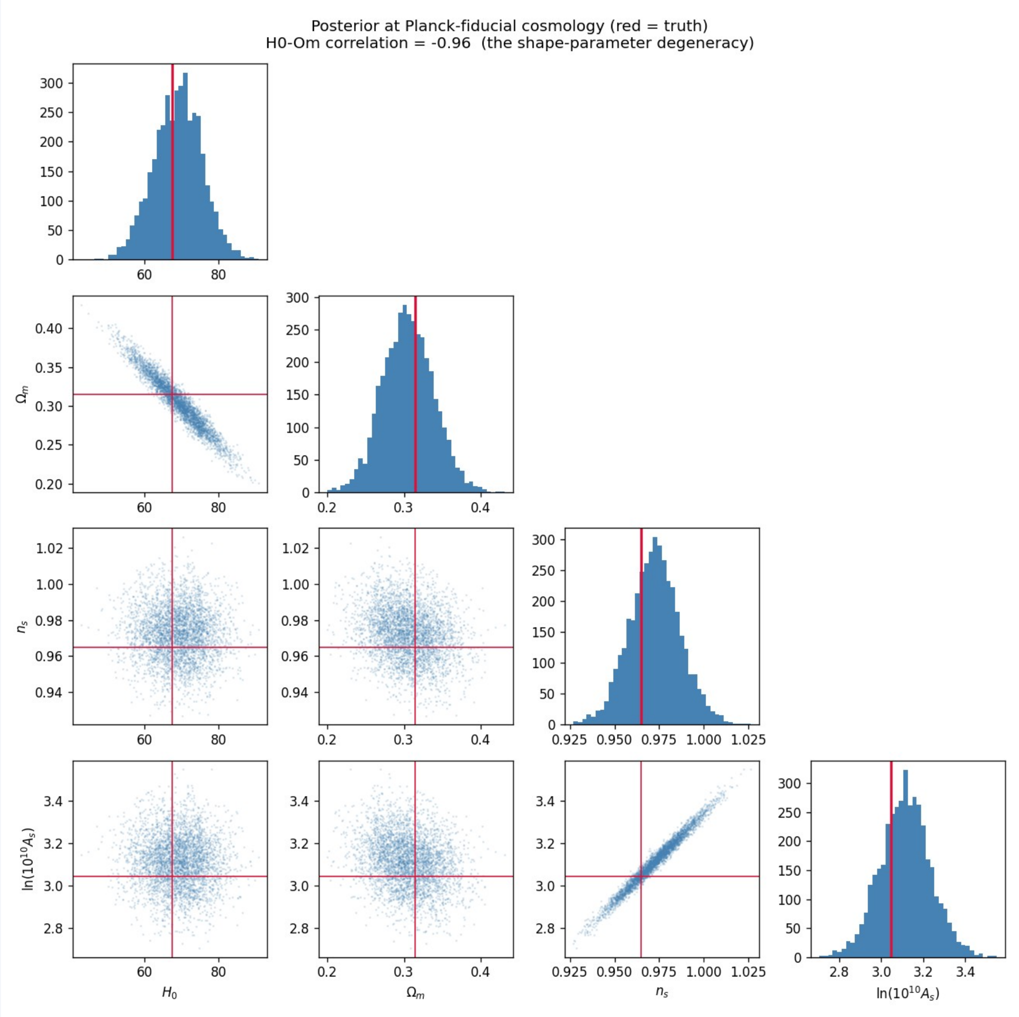


Posterior mean vs true (test set). r = correlation, rmse = error.

Ω_m: r = 0.97 n_s: r = 0.91 A_s: r = 0.87 — strongly recovered.

H_0: r = 0.30 — weakly constrained, as physics predicts (next slide).

Inference at the Planck-2018 cosmology



Posterior corner; red = truth

Truth is recovered:

Ω_m = 0.305 ± 0.033 (true 0.315)

n_s = 0.972 ± 0.014 (true 0.965)

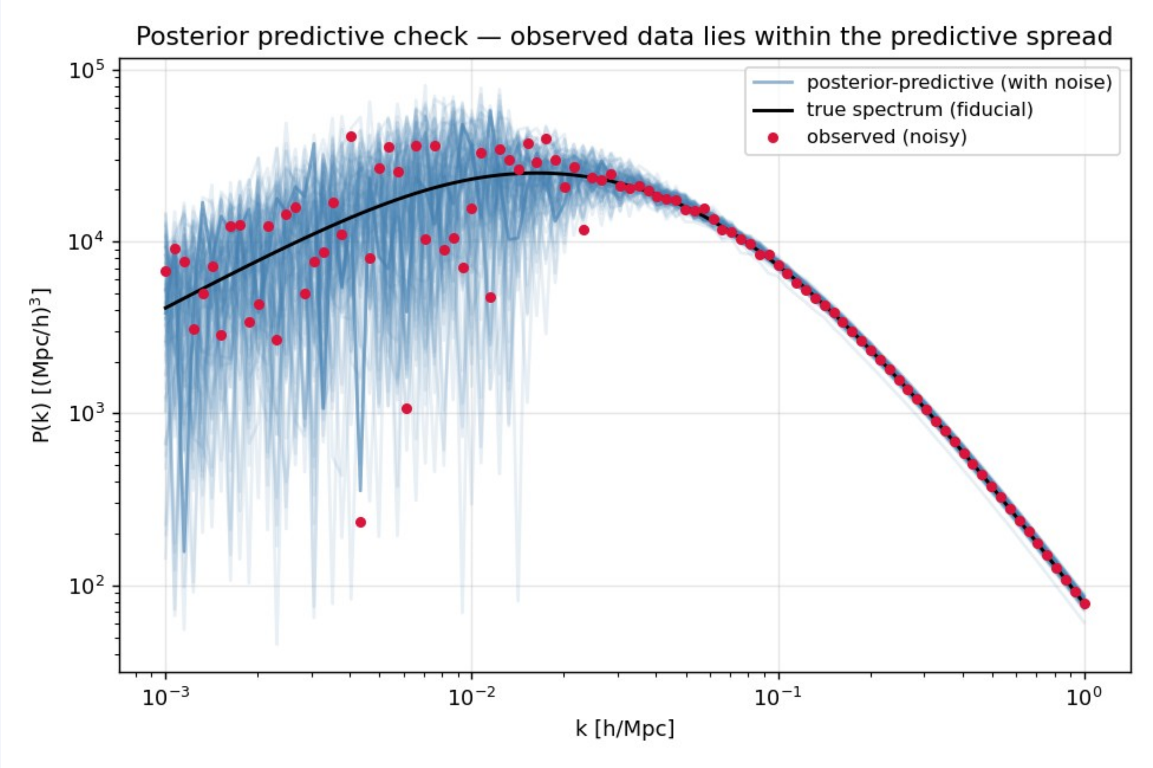
H_0 = 69.0 ± 6.4 (true 67.4) — broad

The H_0–Ω_m degeneracy

Correlation = −0.96. The linear spectrum's shape depends on $\Gamma \approx \Omega_m h$, so H_0 and Ω_m trade off along constant Γ.

A broad, calibrated H_0 is the correct scientific result — not a failure.

Model check & limitations



Posterior predictive spectra bracket the observation

Posterior predictive check passes

Predicted spectra tightly bracket the observed P(k) at all scales.

Honest limitations

- Gaussian posterior → mild tail overconfidence (→ use a normalizing flow / BayesFlow).
- BBKS analytic simulator, linear only → use CAMB; no BAO / nonlinear.
- Idealized diagonal noise; no galaxy bias / RSD / survey window.
- H_0 weak by design (h-unit shape degeneracy).

Take-home

1. We built a calibrated, amortized neural posterior estimator for ΛCDM parameters from the linear P(k).
2. Ω_m, n_s, A_s are well constrained and recovered; SBC + coverage confirm calibration.
3. H_0 is weakly constrained with a strong H_0–Ω_m degeneracy (−0.96) — the correct shape-parameter physics.

Maps the whole course: Monte Carlo · priors · PPC · SBC · likelihood-free · NPE.   Contact: [email]